

Unit 5: Worked Examples

1. Complete the table below for each unit rate.

Distance (Miles)	Time (Hours)	Unit rate (Miles per Hour)
14	1	$\frac{14}{1} = 14$
350	50	$\frac{350}{50} = 7$
112	4	$\frac{112}{4} = 28$

2. Complete the table to show how many ounces of each beverage are needed to make different sized batches of Mark's favorite drink.

Sweet Tea (In Ounces)	Lemonade (in Ounces)
6	4
$42 \div 1\frac{1}{2} \rightarrow$	28
30	$\times 1\frac{1}{2} \leftarrow 20$
$9 \div 1\frac{1}{2} \rightarrow$	6

$\frac{6}{4} \rightarrow \frac{3}{2}$ or $1\frac{1}{2}$ per ounce

3. Adeline made 160 bracelets to sell at the school fundraiser. 60% of the bracelets were sold. How many bracelets were left?

$$160 \times 0.6 = 96 \quad 160 - 96 = 64 \text{ bracelets left}$$

4. A cookbook shows that 16 tablespoons equal 1 cup. If a recipe in this book requires $3\frac{1}{2}$ cups of sugar, how many tablespoons will that be?

$$16 \times 3\frac{1}{2} \rightarrow \frac{16}{1} \times \frac{7}{2} \rightarrow \frac{112}{2} \rightarrow 56 \text{ tablespoons}$$

5. Mai has to fill and seal 70 packages of homemade trail mix to sell at the upcoming farmer's market. She filled and sealed 31 packages in 27 minutes. Mai wonders, "If I keep working at this rate, will I be able to get all of these done in less than 1 hour?"

$$31 \div 27 = \sim 1.149 \text{ per minute}$$

$$\begin{array}{r} \times \text{ 60 minutes} \\ \hline \sim 68.89 \rightarrow 69 \text{ per hour; no she will not get all} \\ \text{done in 1 hr at this} \\ \text{rate} \end{array}$$